

RAG Evaluation — Developer Cheatsheet

Measure, diagnose, and improve Retrieval-Augmented Generation (RAG) performance using the three core evaluation metrics with [TruLens](#).

[!IMPORTANT] Community cheatsheet — not official Snowflake documentation. For the authoritative reference, see [TruLens documentation](#).

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The 3 Core Metrics

Metric	Checks	Nickname
Groundedness	Are all answers based on retrieved data?	Hallucination Detector
Context Relevance	Did we retrieve the right data?	Retriever Health Check
Answer Relevance	Did we answer the actual question?	Response Quality Check

1. Groundedness

Measures whether the generated answer is factually supported by the retrieved context — no hallucinations or made-up facts.

Example

Context retrieved:

“The Eiffel Tower was completed in 1889 and stands 330 meters tall.”

Good (grounded):

“The Eiffel Tower is 330 meters tall.”

Bad (not grounded):

“The Eiffel Tower is 330 meters tall and was designed by Gustave Eiffel in collaboration with Leonardo da Vinci.”

Why bad: hallucinates information not supported by the context.

2. Context Relevance

Measures whether the retrieved documents or chunks actually contain information relevant to answering the user’s question.

Example

User question:

“What are the health benefits of green tea?”

Good (relevant context):

- “Green tea contains antioxidants called catechins that may reduce inflammation.”
- “Studies show green tea consumption is associated with lower cardiovascular disease risk.”

Bad (irrelevant context):

- “The tea ceremony in Japan dates back to the 9th century.”
- “Green tea is grown primarily in China, Japan, and India.”

Why bad: these contexts don’t help answer the question about health benefits.

3. Answer Relevance

Measures whether the generated answer actually addresses the user's question clearly and directly.

Example

User question:

| "How do I reset my password?"

Good (relevant answer):

| "To reset your password, click 'Forgot Password' on the login page, enter your email, and follow the link sent to your inbox."

Bad (irrelevant answer):

| "Passwords are important for security. Strong passwords should contain uppercase, lowercase, numbers, and symbols."

Why bad: talks about passwords generally but doesn't answer how to reset it.

How the Three Metrics Work Together

graph TD

```
A[User Query] --> B[Retrieval: Context Relevance]
B --> C[Generation: Groundedness]
C --> D[Final Answer: Answer Relevance]
```

Each stage builds on the previous:

- **Context Relevance:** Did we get the right data?
- **Groundedness:** Is the answer supported by that data?
- **Answer Relevance:** Does the response actually answer the question?

End-to-End Example

Query:

| "What is the refund policy for online orders?"

1. **Retrieval:** "Online orders can be returned within 30 days for a full refund." — High Context Relevance

2. **Generation:** “You can get a full refund if you return within 30 days.” — High Groundedness
3. **Final Answer:** Directly answers the refund question. — High Answer Relevance

All three metrics high = high-quality RAG response.

Fix Order

[!NOTE] Bad retrieval breaks everything downstream. Fix in order: Context Relevance first, then Groundedness, then Answer Relevance.

Good RAG = Right Data (Context Relevance) + No Hallucinations (Groundedness) + On Topic (Answer Relevance).

Scoring Quick View

Metric	Ship It	Tune It	Fix It	Stop
Groundedness	0.90–1.0	0.70–0.89	0.50–0.69	< 0.50
Context Relevance	0.80–1.0	0.60–0.79	0.40–0.59	< 0.40
Answer Relevance	0.90–1.0	0.70–0.89	0.50–0.69	< 0.50

[!NOTE] Thresholds should be set case-by-case based on your application’s goals and tolerance for error.

Quick Wins

If Groundedness is low: Add to prompt: “*Base your answer ONLY on the provided context. If unsure, say ‘I don’t have enough information.’*”

If Context Relevance is low: Improve retrieval configuration, embedding quality, or metadata filtering to return more relevant context.

If Answer Relevance is low: Add to prompt: “*Answer the question directly and concisely. Stay focused on what was asked.*”

Rules of Thumb

1. **You can’t improve what you don’t measure.**
2. **Fix retrieval first** — garbage in = garbage out.
3. **All three must be high** for reliable RAG evaluation.

4. **Monitor continuously** — data and model drift can affect scores.

Common Score Patterns

Pattern	Meaning	Fix
All high	Ready for production	Deploy and monitor
All low	System broken	Debug fundamentals
Low context	Retrieval issue	Tune embeddings or search
Low groundedness	Hallucination risk	Add stricter prompts
Low answer	Prompting issue	Simplify and focus

Readiness Checklist

- ☐ Metrics meet internal thresholds
- ☐ Tested on diverse queries
- ☐ Monitoring and alerts in place
- ☐ Escalation path defined
- ☐ Baseline metrics tracked

References

- [TruLens Documentation](#)
- [TruLens Home](#)
- [Snowflake Cortex Search](#)
- [RAG with Snowflake Cortex \(Tutorial 2\)](#)